Problem Set XI, Macroeconomics III
University of Copenhagen

Exercise 1

The consumer's problem is given by

$$\max_{\{C_t, A_{t+1}\}_t^\infty} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t u(C_t)$$
$$\text{subject to} \quad \frac{A_{t+1}}{P_t} + C_t = \frac{W_t}{P_t} + (1 + i_t)\frac{A_t}{P_t}$$

where

$$u(c_t) = \frac{c_t^{1-\sigma} - 1}{1 - \sigma}$$

C_t and A_t represent consumption and savings, respectively. The nominal wage is represented by W_t, A_t is a nominal bond, P_t is the price level.

a) Set up the Lagrangian for this problem and take the first order conditions with respect to consumption and savings. When combining them, which (hopefully) familiar equation do you obtain? What happens to consumption in period t if i_{t+1} rises, all else equal? What happens to consumption growth?

b) Impose market clearing, such that $C_t = Y_t$. Take logs, such that $\log(Y_t) = y_t$. Show that the first order conditions imply $y_t = \mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma}(\mathbb{E}_t[r_{t+1}] - \rho)$, where $r_t = i_t - \pi_t$. Explain the economic significance of σ. What does it represent? *(Assume that the logarithm of an expectation is approximately equal to the expectation of the logarithm:* $\log(\mathbb{E}[x]) \approx \mathbb{E}[\log(x)]$. *Recall that* $\log(1 + x) \approx x$.*)*

Recall the other two equations that describe the New Keynesian model from the lecture:

$$\begin{aligned}\pi_t &= \kappa y_t + \beta \mathbb{E}_t[\pi_{t+1}] \\ r_{t+1} &= \rho + \psi_y \mathbb{E}_t[y_{t+1}] + \psi_\pi \mathbb{E}_t[\pi_{t+1}] + \varepsilon_t \\ \text{with } \kappa &= \frac{\alpha(\phi-1)(1-\beta(1-\alpha))}{1-\alpha}; \quad \phi > 1, \beta \in (0,1), \alpha \in \{0,1\}\end{aligned}$$

The monetary policy authority surprises the economy, setting $\varepsilon_t = 1$, but it commits to all future shocks being zero; $\varepsilon_{>1} = 0$. In period $t < 0$, all variables are normalized to zero, i.e., $y = \pi = r = 0$.

c) Explain why we only need to focus on period 0 when investigating the impact of the shock described above. Why does it not have long-lasting effects?

d) Show how $\varepsilon_0 = 1$ affects y_0 and π_0.

e) For each of the variables ϕ, α and β, explain how they alter the effect of monetary policy on output and inflation. Give intuitive explanations for each.

Exercise 2

Consider the Calvo economy studied in lectures where each of a continuum of producers' ideal (log) price in each period is given by:

$$p_t^\star = \phi m_t + (1-\phi) p_t.$$

Here, ϕ is an inverse measure of the degree of **real rigidity** in the economy. In each period, firms are only able to change prices with probability $\alpha \in (0,1)$. Assume that the household discount factor is $\beta \in (0,1)$, and that the economy is in long-run equilibrium with all (log) prices equal to m, which we normalize to zero. Assume in period 1 there's a one-time **permanent** increase in m to $m_1 > m$ (an MIT shock). Conjecture that the behavior of the price level for $t \geq 1$ is described by:

$$p_t = \left(1 - \lambda^t\right) m_1.$$

a) Explain why this conjecture is or is not reasonable.

b) Find λ in terms of the primitive parameters, α, β, and ϕ. *(Hint: Try to solve for y_1 and π_1 as a function of λ and m_1; then, find the expectation of future inflation and use the NKPC to pin down λ. There is no need for an infinite forward substitution.*)

c) How do increases in each of the parameters α and ϕ affect λ? Explain your answer.

d) What does your answer in c) tell you about the propagation of monetary policy shocks? When are effects large and when are they small?

You will also need some subset of the following equations, which detail the economy:

$$y_t = m_t - p_t \qquad \text{(Output)}$$

$$\pi_t = p_t - p_{t-1} \qquad \text{(Inflation)}$$

$$p_t = \alpha x_t + (1-\alpha) p_{t-1} \qquad \text{(Price level)}$$

$$\pi_t = \beta \mathbb{E}_t \left[\pi_{t+1}\right] + \kappa y_t, \quad \kappa = \frac{\alpha\phi\left[1-\beta(1-\alpha)\right]}{1-\alpha}. \qquad \text{(NKPC)}$$

Recall that $p_t^\star$ differ from optimal prices x_t, which are described in lectures.